Lecture 4: Change of Measure

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1 Risk-neutral Measure

We have seen that, under no arbitrage, there is a unique risk-neutral measure in a N-period binomial market. Let us denote by $\tilde{p}$ the risk-neutral probability[1], and by $\tilde{P}$ the probability measure induced by $\tilde{p}$. Then for any $\omega \in \Omega$,

$$\tilde{P}(\text{the } i\text{-th letter of } \omega \text{ is } H) = \tilde{p}, \text{ for any } 1 \leq i \leq N.$$

As usual, we denote by $\tilde{E}$ the expectation under $\tilde{P}$. Then a security that pays $V_N(\omega)$ at its maturity N, can be priced at any time $0 \leq k \leq N$: the fair value at time k is[2]

$$V_k = \frac{1}{(1+r)^{N-k}} \tilde{E}(V_N | \mathcal{F}_k). \tag{1}$$

In particular, the time zero price is

$$V_0 = \frac{1}{(1+r)^N} \tilde{E}(V_N | \mathcal{F}_0) = \frac{1}{(1+r)^N} \tilde{E}(V_N). \tag{2}$$

Because there is only one payment at maturity, one can use (2) to easily compute the fair value at time zero with just one go (instead of using backward induction):

$$V_0 = \frac{1}{(1+r)^N} \sum_{\omega \in \Omega} V_N(\omega) \tilde{P}(\omega). \tag{3}$$

Because we only consider finite probability space here, the summation in (3) is finite. Similarly, one can use (1) to compute the time k price with conditional probabilities, but the corresponding summation will involve few terms because we now have more information available to us (so less randomness).

The risk-neutral measure is the right measure to work with if one wants to price derivatives. But it is also possible to use the actual measure for pricing if he is careful enough.

2 Change of Measure

We begin by considering a general finite probability space $(\Omega, \mathcal{F}, P)$. On this space, we have another probability measure $\tilde{P}$. Let us assume that P and $\tilde{P}$ both give positive probability to every element of Ω, so that we can form the quotient

$$Z(\omega) = \frac{\tilde{P}(\omega)}{P(\omega)}. \tag{4}$$

Clearly, Z is a positive function on Ω, Z is a positive random variable. And one can compute the expectation of Z under measure P:

$$E(Z) = \sum_{\omega \in \Omega} Z(\omega) P(\omega) = \sum_{\omega \in \Omega} \frac{\tilde{P}(\omega)}{P(\omega)} \cdot P(\omega) = \sum_{\omega \in \Omega} \tilde{P}(\omega) = 1. \tag{5}$$

[1] That is, $\tilde{p} = \frac{1+r-d}{u-d}$.

[2] You may have noticed that the fair value is just the risk-neutral expectation of discounted future cash flow. This is also true even if there are multiple payments before maturity.

In general, for any random variable Y,

$$E(ZY) \;=\; \sum_{\omega\in\Omega} Z(\omega)Y(\omega)P(\omega) \;=\; \sum_{\omega\in\Omega} \frac{\tilde{P}(\omega)}{P(\omega)} \cdot Y(\omega)P(\omega) \;=\; \sum_{\omega\in\Omega} Y(\omega)\tilde{P}(\omega) \;=\; \tilde{E}(Y). \tag{6}$$

In view of (6), the random variable Z is called the **Radon-Nikodým derivative** of $\tilde{P}$ with respect to P.

Returning to the N-period binomial model with constant parameters p, u, d, r, we also have two probability measures: the actual measure induced by p, and the risk-neutral measure induced by $\tilde{p}$. We can easily find the Radon-Nykodým derivative of the risk-neutral measure w.r.t. the actual measure. We illustrate this using an example.

Example 1. In a three-period binomial model, $\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$. Using (6) we obtain that

$$\begin{aligned}
Z(HHH) &= \left(\frac{\tilde{p}}{p}\right)^3, \\
Z(HHT) &= Z(HTH) \;=\; Z(THH) \;=\; \left(\frac{\tilde{p}}{p}\right)^2 \cdot \frac{\tilde{q}}{q}, \\
Z(TTH) &= Z(THT) \;=\; Z(HTT) \;=\; \frac{\tilde{p}}{p} \cdot \left(\frac{\tilde{q}}{q}\right)^2, \\
Z(TTT) &= \left(\frac{\tilde{q}}{q}\right)^3.
\end{aligned}$$

In general, in a N-period binomial model, for any $\omega \in \Omega$, it is easy to verify that

$$Z(\omega) \;=\; \left(\frac{\tilde{p}}{p}\right)^{\#H(\omega)} \cdot \left(\frac{\tilde{q}}{q}\right)^{\#T(\omega)}, \tag{7}$$

where $\#H(\omega)$ is the number of H's in string ω, and $\#T(\omega)$ is the number of T's in string ω. In order to price derivative securities under the actual measure, we define the **state price density random variable** as

$$\zeta(\omega) \;=\; \frac{Z(\omega)}{(1+r)^N}. \tag{8}$$

The role that ζ plays can be seen from the following equation: for any derivative with payoff V_N at time N, the time zero price of it is

$$V_0 \;=\; \frac{1}{(1+r)^N}\tilde{E}(V_N) \;=\; \frac{1}{(1+r)^N}E(ZV_N) \;=\; E(\zeta V_N) \;=\; \sum_{\omega\in\Omega} \zeta(\omega)V_N(\omega)P(\omega). \tag{9}$$

3 Radon-Nykodým Derivative Process

The Radon-Nykodým derivative Z in (4) relates $\tilde{P}$ to P, as a result, pricing under the actual measure P at time zero is possible. Similarly, the time k price of a security can also be computed under the actual measure. To see this, we need a lemma.

Lemma 1. *Let Y be a random variable measurable with respect to $\mathcal{F}_k$. That is, the value of Y only depends on the first k coin tosses. Then*

$$\tilde{E}(Y) = E(YZ_k), \;\; \textit{where } Z_k = E(Z|\mathcal{F}_k). \tag{10}$$

Proof. Using (6) we have

$$\tilde{E}(Y) \;=\; E(ZY) \;=\; E(E(ZY|\mathcal{F}_k)) \;=\; E(YE(Z|\mathcal{F}_k)) \;=\; E(YZ_k). \tag{11}$$

This completes the proof. □

Note that (10) is a generalization of (6), if we know the random variable Y is actually $\mathcal{F}_k$-measurable. Moreover, one can easily obtain that, if the result of the first k coin tosses is $\overline{\omega}_1 \dots \overline{\omega}_k$, then define a $\mathcal{F}_k$-measurable random variable

$$Y(\omega_1 \dots \omega_k \omega_{k+1} \dots \omega_N) \;=\; \mathbb{I}_{\{\overline{\omega}_1 \dots \overline{\omega}_k\}}(\omega_1 \dots \omega_k \omega_{k+1} \dots \omega_N). \tag{12}$$

That is, Y takes 1 if the first k coin tosses result in $\overline{\omega}_1 \dots \overline{\omega}_k$, otherwise Y takes 0. Using (10) we have

$$\begin{aligned}
\tilde{p}^{\#H(\overline{\omega}_1 \dots \overline{\omega}_k)} \cdot \tilde{q}^{\#T(\overline{\omega}_1 \dots \overline{\omega}_k)} &= \tilde{E}(Y) \;=\; E(Z_k Y) \\
&= Z_k(\overline{\omega}_1 \dots \overline{\omega}_k) p^{\#H(\overline{\omega}_1 \dots \overline{\omega}_k)} q^{\#T(\overline{\omega}_1 \dots \overline{\omega}_k)},
\end{aligned} \tag{13}$$

where $\#H(\overline{\omega}_1 \dots \overline{\omega}_k)$ is the number of H's in string $\overline{\omega}_1 \dots \overline{\omega}_k$, $\#T(\overline{\omega}_1 \dots \overline{\omega}_k)$ is the number of T's in string $\overline{\omega}_1 \dots \overline{\omega}_k$. Thus,

$$Z_k(\overline{\omega}_1 \dots \overline{\omega}_k) \;=\; \left(\frac{\tilde{p}}{p}\right)^{\#H(\overline{\omega}_1 \dots \overline{\omega}_k)} \cdot \left(\frac{\tilde{q}}{q}\right)^{\#T(\overline{\omega}_1 \dots \overline{\omega}_k)}. \tag{14}$$

Clearly, we have $Z_N = Z$.

Remark 1. *The stochastic process $\{Z_k\}$ is adapted to the filtration $\{\mathcal{F}_k\}$. Lemma 1 says, to convert the risk-neutral expectation of a $\mathcal{F}_k$-measurable random variable Y to the expectation under the actual measure, we just need to use Z_k.*

The conditional expectation version of Lemma 1 is the following:

Lemma 2. *Assume that $0 \le n \le m \le N$, and Y is a $\mathcal{F}_m$-measurable random variable. Then*

$$\tilde{E}(Y|\mathcal{F}_n) \;=\; \frac{1}{Z_n} E(Z_m Y|\mathcal{F}_n). \tag{15}$$

Proof. For any strings of $\{H, T\}$ with length k, $\omega_1 \dots \omega_k$,

$$\begin{aligned}
\tilde{E}(Y|\mathcal{F}_n) &= \sum_{\omega_{n+1},\dots,\omega_m} Y(\omega_1 \dots \omega_m) \cdot \tilde{p}^{\#H(\omega_{n+1} \dots \omega_m)} \cdot \tilde{q}^{\#T(\omega_{n+1} \dots \omega_m)} \\
&= \left(\frac{\tilde{p}}{p}\right)^{-\#H(\omega_1 \dots \omega_n)} \cdot \left(\frac{\tilde{q}}{q}\right)^{-\#T(\omega_1 \dots \omega_n)} \sum_{\omega_{n+1},\dots,\omega_m} Y(\omega_1 \dots \omega_m) \cdot \left(\frac{\tilde{p}}{p}\right)^{\#H(\omega_1 \dots \omega_m)} \cdot \left(\frac{\tilde{q}}{q}\right)^{\#T(\omega_1 \dots \omega_m)} \\
&\quad \times p^{\#H(\omega_{n+1} \dots \omega_m)} \cdot q^{\#T(\omega_{n+1} \dots \omega_m)} \\
&= \frac{1}{Z_n}(\omega_1 \dots \omega_n) \sum_{\omega_{n+1},\dots,\omega_m} Y(\omega_1 \dots \omega_m) \cdot Z(\omega_1 \dots \omega_m) \cdot p^{\#H(\omega_{n+1} \dots \omega_m)} \cdot q^{\#T(\omega_{n+1} \dots \omega_m)} \\
&= \frac{1}{Z_n} E(Z_m Y|\mathcal{F}_n).
\end{aligned} \tag{16}$$

□

Using Lemma 2 we can price a derivative security that pays off V_N at time N: at time k, the fair price is

$$V_k \;=\; \frac{1}{(1+r)^{N-k}} \tilde{E}(V_N|\mathcal{F}_k) \;=\; \frac{1}{(1+r)^{N-k}} \cdot \frac{1}{Z_k} E(Z_N V_N|\mathcal{F}_k) \;=\; \frac{1}{\zeta_k} E(\zeta_N V_N|\mathcal{F}_k), \tag{17}$$

provided that

$$\zeta_k \;=\; \frac{Z_k}{(1+r)^k}. \tag{18}$$